

the IT infrastructure. MQTT connects devices and networks with applications and middleware using a hub-and-spoke architecture. MQTT protocols provide efficient information routing functions to small, cheap, low-memory and power consuming devices in vulnerable and low bandwidth based networks.

XMPP: This is a communication IoT protocol for message-oriented middleware based on the XML language. It enables the real-time exchange of structured yet extensible data between any two or more network entities. XMPP enables messaging applications to attain authentication, access control, and end-to-end encryption.

AMQP: AMQP is an application layer protocol that supports reliable communication via message delivery assurance primitives like ‘at-most once’, ‘at least once’ and ‘exactly once’ delivery. The AMQP protocol consists of a set of components that route and store messages within a broker service. This protocol enables client applications to talk to the broker, and interact with the AMQP model.

Data Distribution Service (DDS): This is used for real-time machine-to-machine communication. It enables scalable, real-time, dependable, high-performance and interoperable data exchange via the publish-subscribe methodology. DDS can be deployed in platforms ranging from low-footprint devices to the cloud, and supports efficient bandwidth usage as well as the agile orchestration of system components.

Other open source connecting protocols are VSCP, STOMP, etc.

Open source infrastructure

The following are some of the key open source cloud platforms.

OpenStack: This is open source software for creating private and public clouds, built and disseminated by a large and democratic community of developers in collaboration with users. OpenStack is mostly deployed as Infrastructure-as-a-Service (IaaS), whereby virtual servers and other resources are made available to customers.

Cloud Foundry: This is an open Platform-as-a-Service (PaaS), which provides a choice of clouds, developer frameworks and application services. Cloud Foundry makes it faster and easier to build, test, deploy and scale applications.

OpenShift: This is Red Hat’s cloud computing PaaS offering. It is an application platform in the cloud, where app developers and teams can build, test, deploy and run their applications.

Cloudify: Cloudify was developed and designed on the principles of openness to power the IT transformation revolution. It enables organisations to design, build and deliver various business applications and network services.

Other open source cloud platforms are WSO2, Cloud Stack, etc.

Open source hardware

Open source hardware consists of physical artifacts of technology designed and offered by the open design movement.

Arduino: This operates through an array of hardware specifications that can be given to interactive electronics. The software of Arduino comes in the form of the Arduino programming language and Integrated Development Environment (IDE).

Raspberry Pi: This inexpensive computer runs Linux, and also provides a set of GPIO (general purpose input/output) pins that allow you to

control electronic components for physical computing and explore the Internet of Things (IoT).

mangOH: This industrial grade open source hardware is designed to address common IoT pain points and deliver out-of-the-box IoT applications as prototypes, helping to bring IoT products to the market sooner.

Open source operating systems

Android Things (formerly known as Brillo): This operating system from Google is built on the power of Android for embedded devices. It has access to all the Android APIs, Google services and developer tools.

RIOT (Real-time IoT): This is a real-time operating system, which is based on the micro-kernel architecture for IoT. RIOT has been developed by a consortium of universities in Germany and France.

Zephyr: This is another real-time operating system (RTOS), similar to RIOT, launched by the Linux Foundation and available under the Apache license. It is small, scalable and optimised for resource constrained devices.

Ubuntu Core: Ubuntu Core is based on the Linux OS developed for IoT devices. It is delivered as Linux application packages called Snaps.

In conclusion, open source tools used in IoT enable efficiency in data delivery, and help to manage highly secure, reliable and scalable IoT solutions. 

By: Dr Gopala Krishna Behara and Raju Myadam

Dr Gopala Krishna Behara is a lead enterprise architect in the digital architecture division of Wipro. He has 24 years of experience in the IT industry.

Raju Myadam is a chief enterprise architect in the digital architecture division of Wipro. He has 21 years of experience in the IT industry.

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